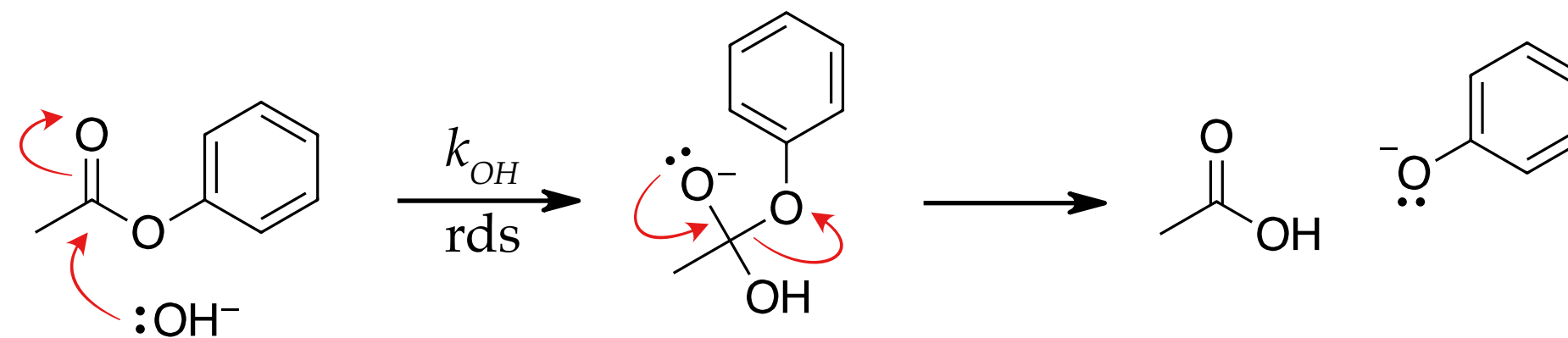
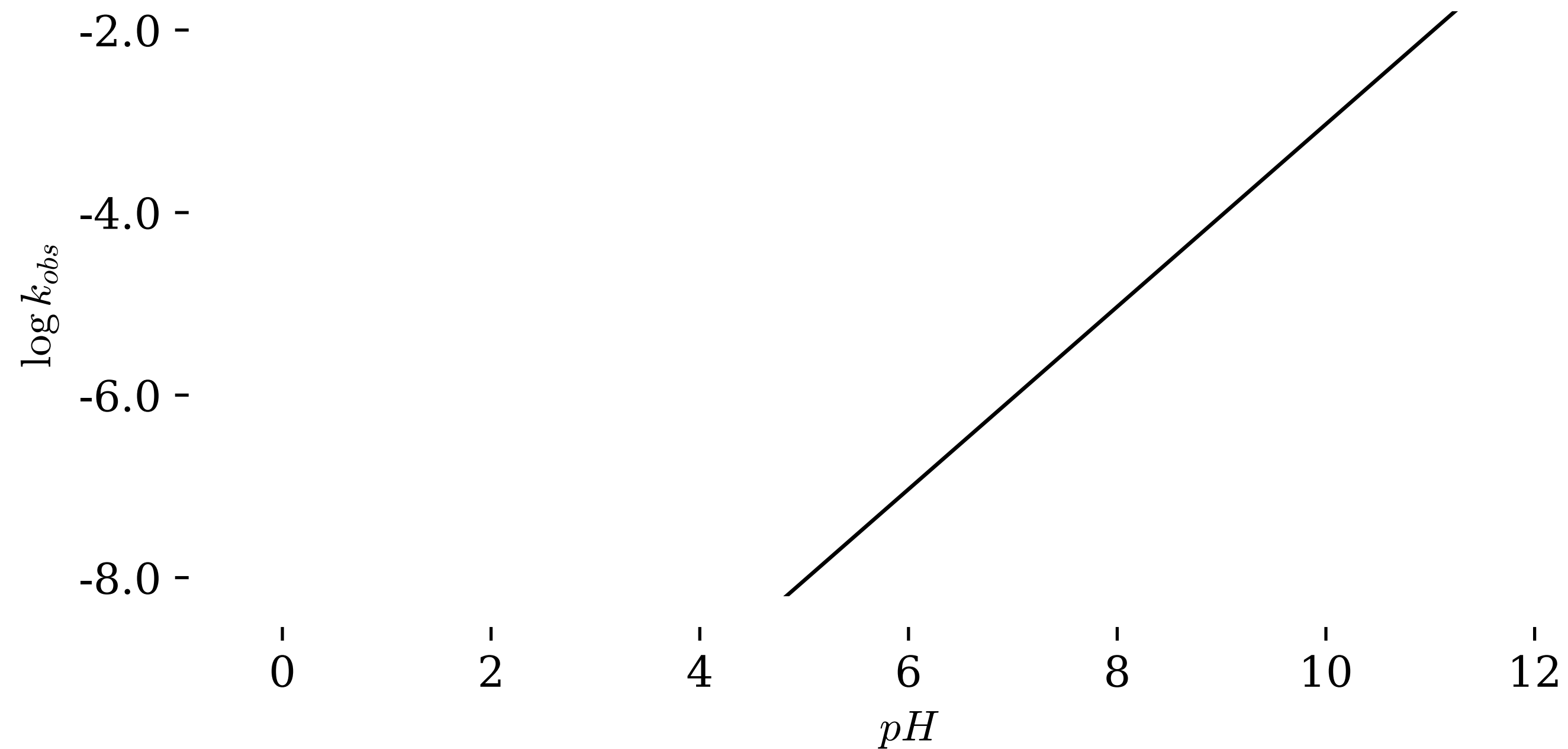


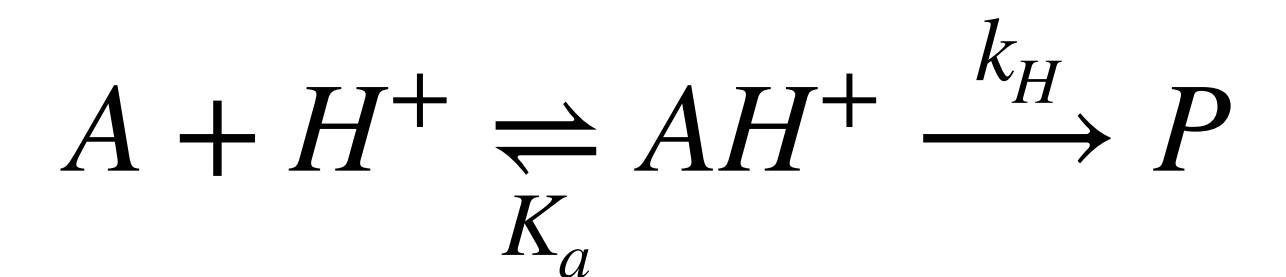
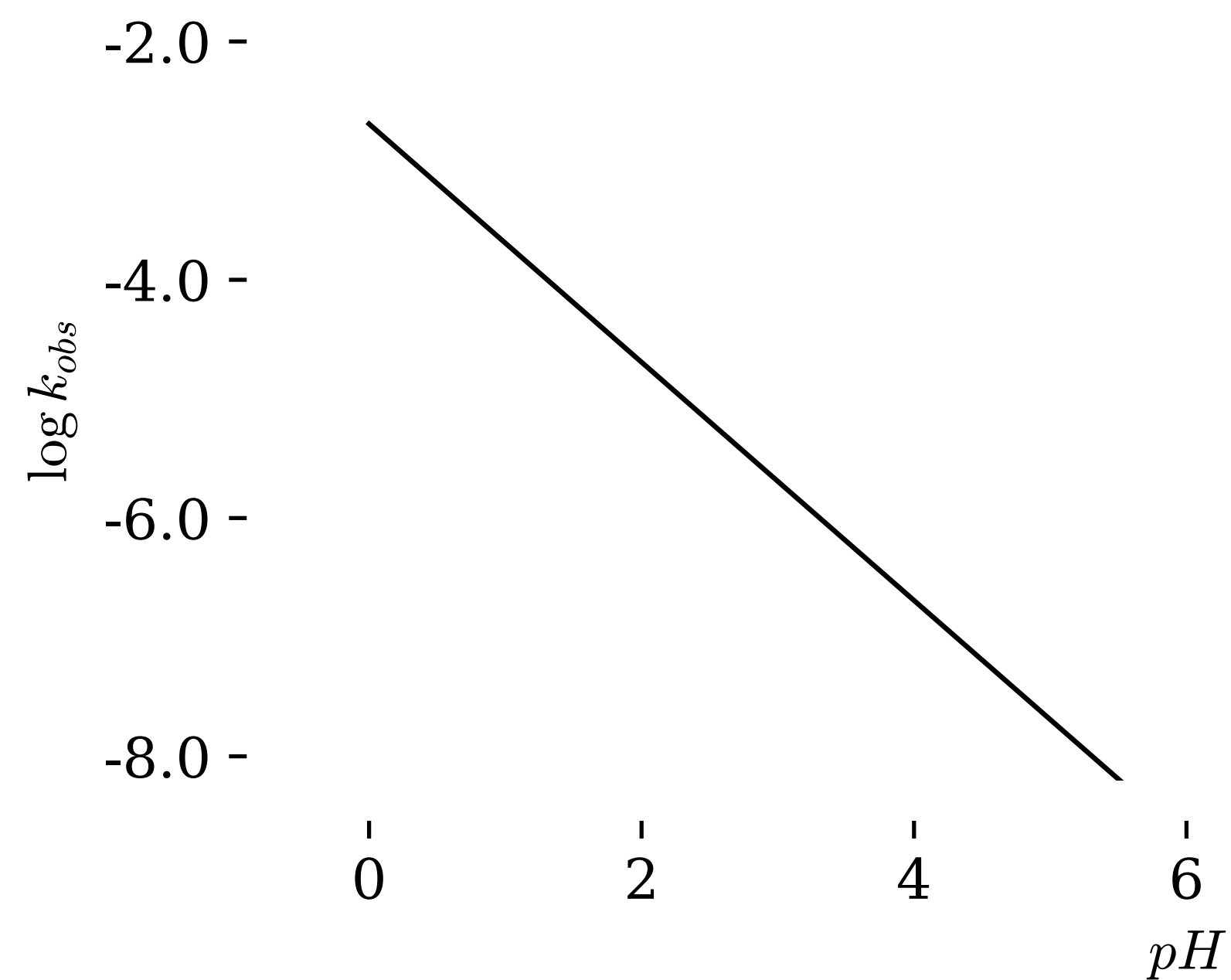
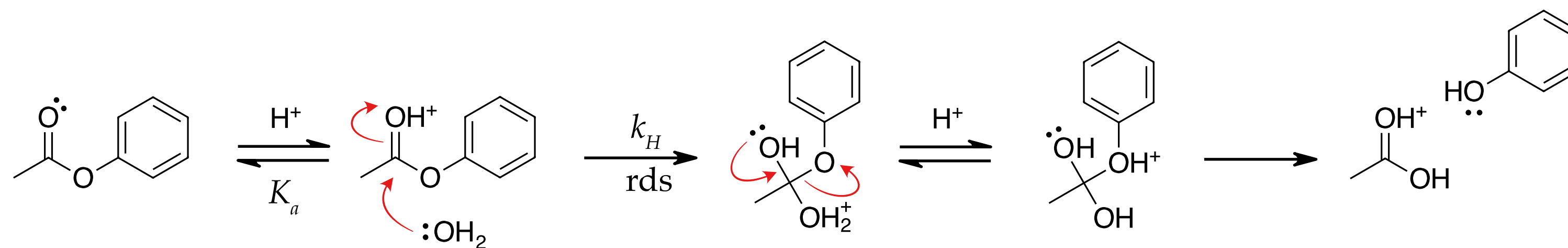
Specific Base



$$\frac{d}{dt} [P] = k_{OH^-} \frac{K_w}{[H^+]} [A]_t$$



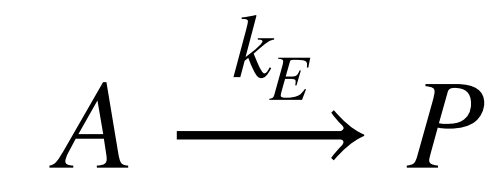
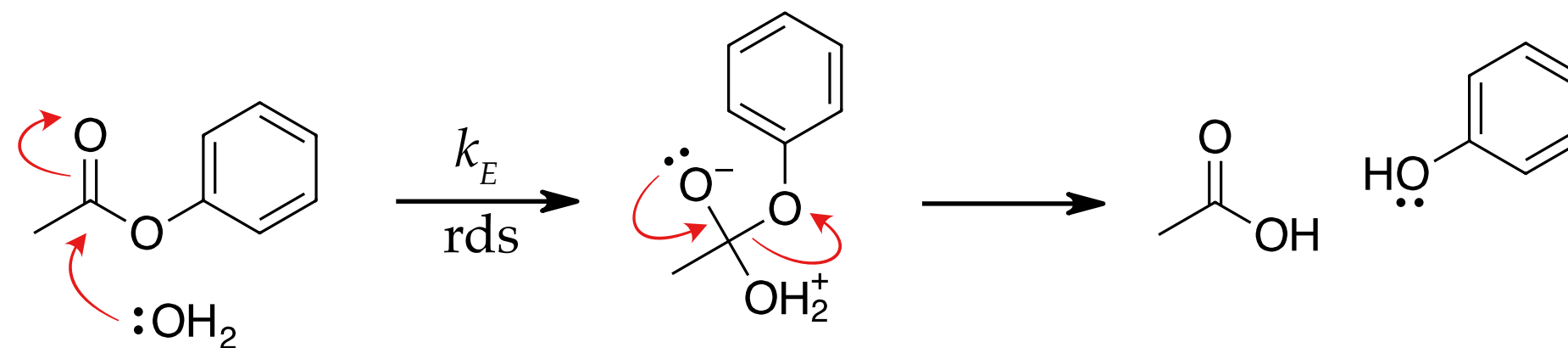
Specific Acid



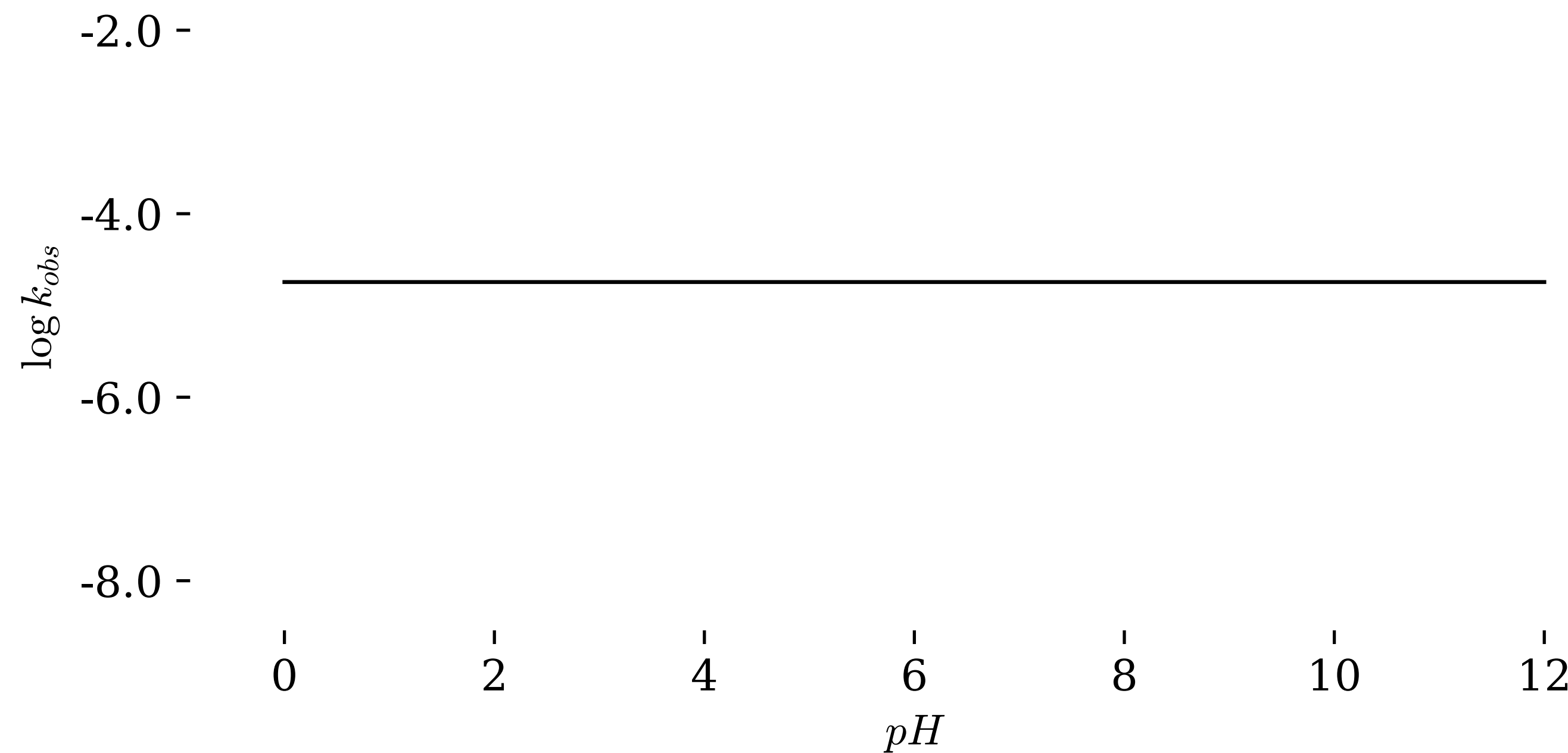
$$\frac{d}{dt} [P] = k_H \frac{[H^+]}{K_a + [H^+]} [A]_t$$

$$\frac{d}{dt} [P] = \frac{k_H}{K_a} [H^+] [A]_t$$

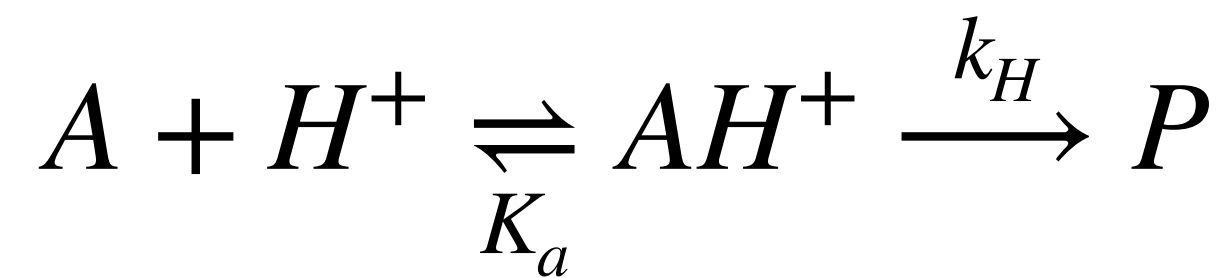
No Acid, No Base



$$\frac{d}{dt} [P] = k_E [A]_t$$



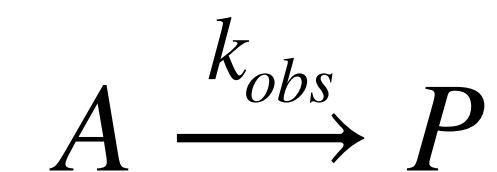
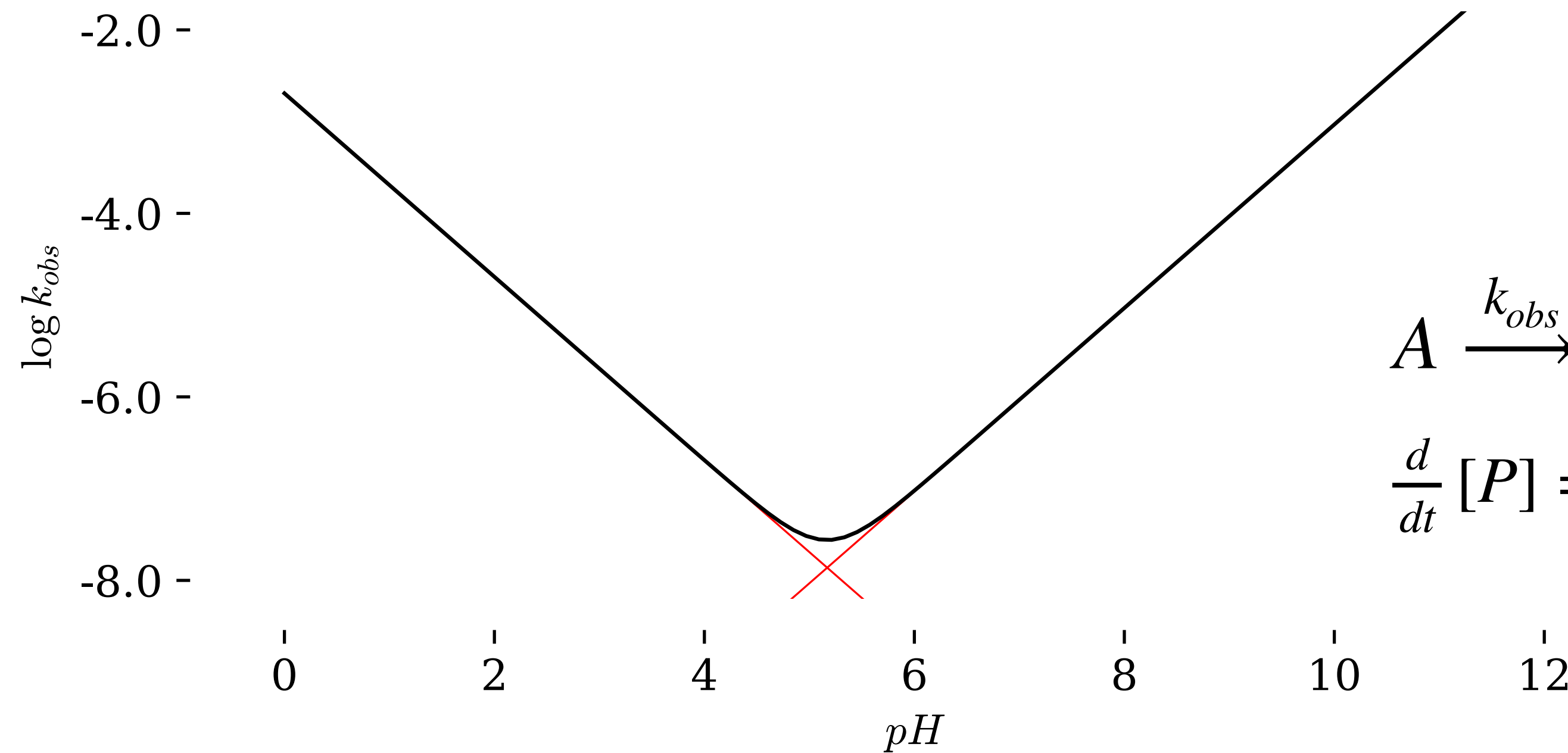
Specific Acid & Base



$$\frac{d}{dt} [P] = \frac{k_H}{K_a} [H^+] [A]_t$$

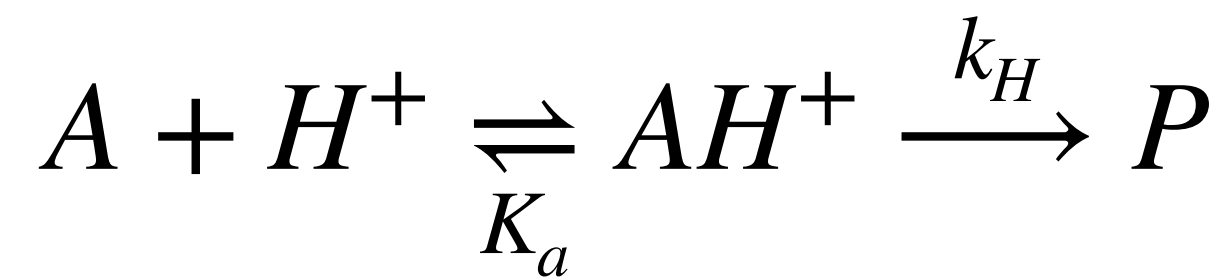


$$\frac{d}{dt} [P] = k_{OH} \frac{K_w}{[H^+]} [A]_t$$



$$\frac{d}{dt} [P] = \left(\frac{k_H}{K_a} [H^+] + k_{OH} \frac{K_w}{[H^+]} \right) [A]_t$$

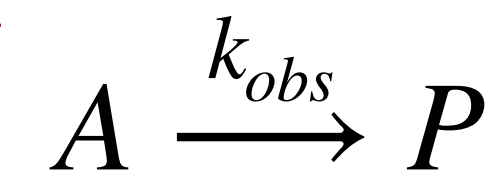
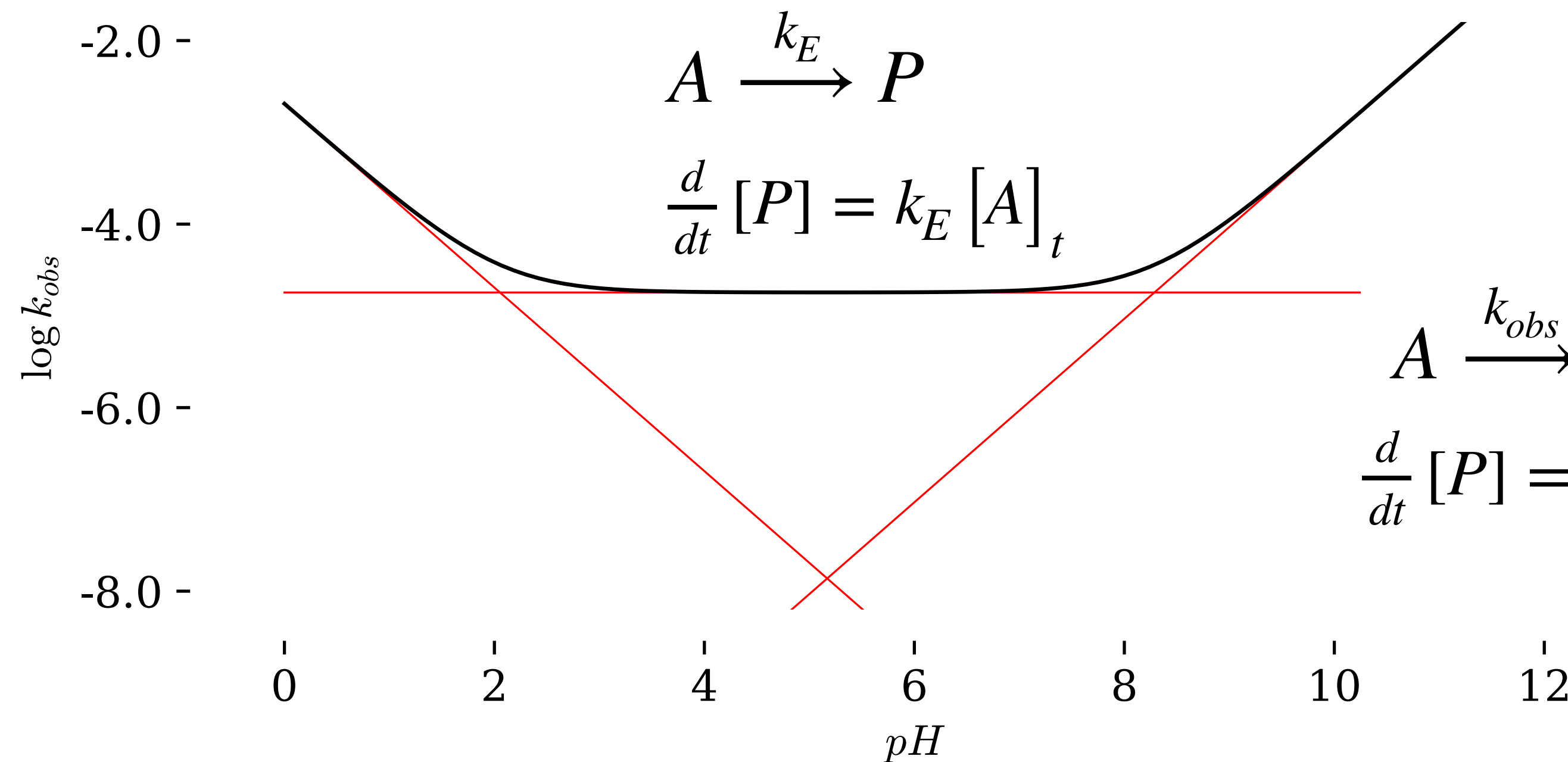
Specific Acid & Base and Water



$$\frac{d}{dt} [P] = \frac{k_H}{K_a} [H^+] [A]_t$$

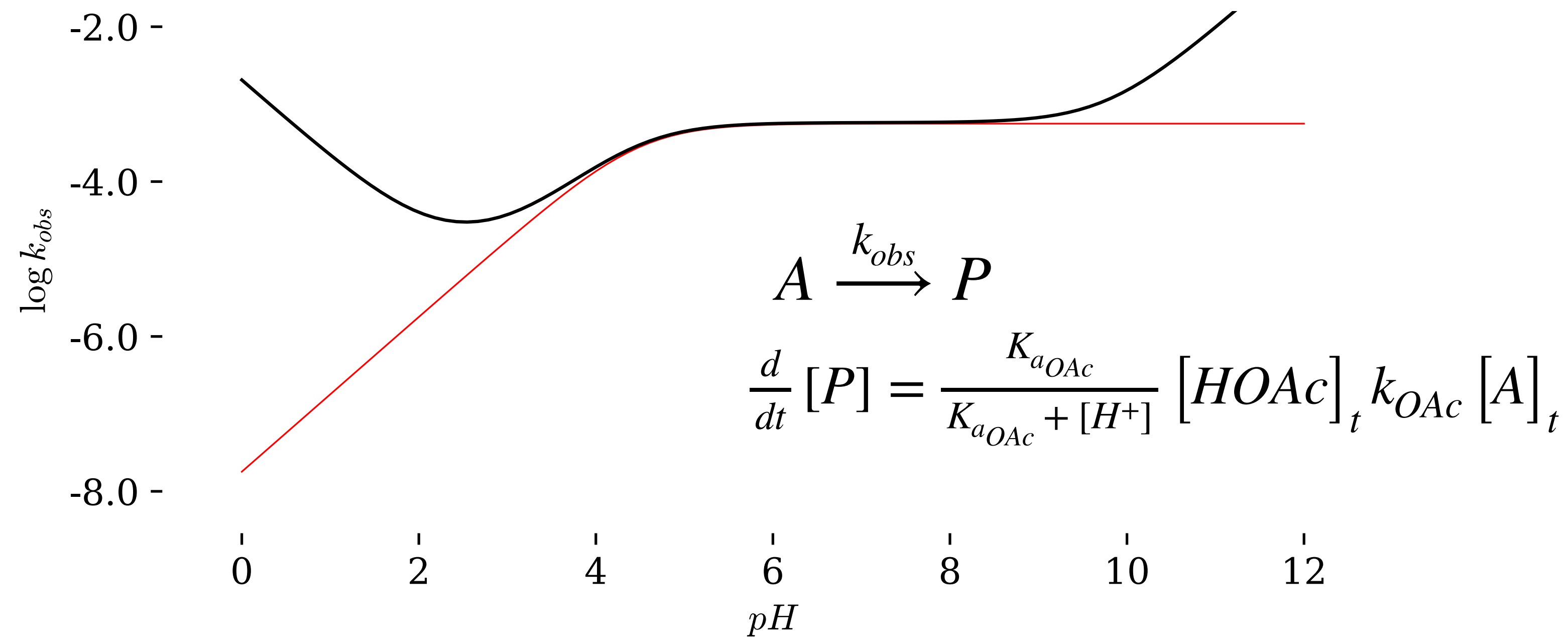
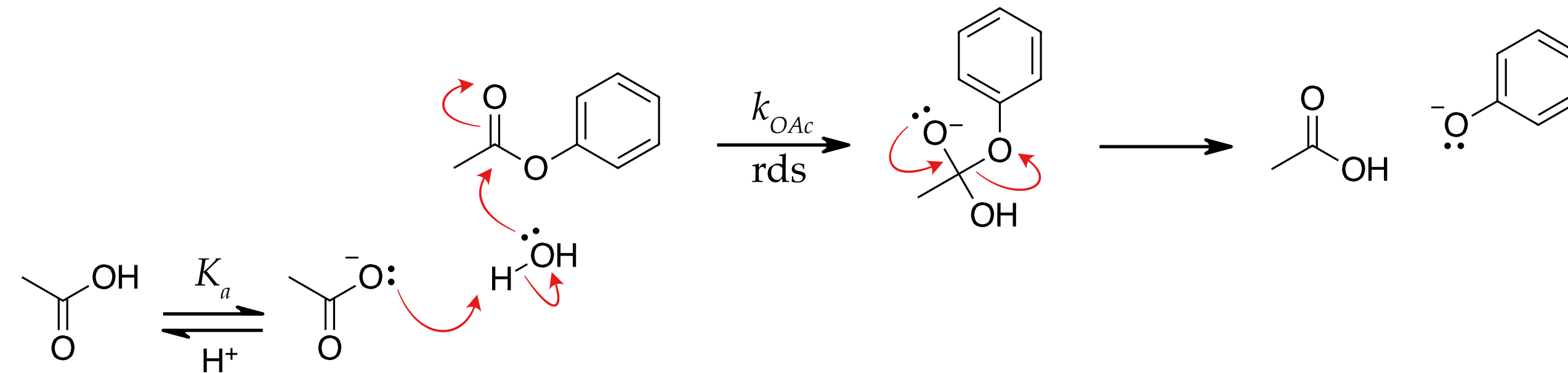


$$\frac{d}{dt} [P] = k_{OH} \frac{K_w}{[H^+]} [A]_t$$

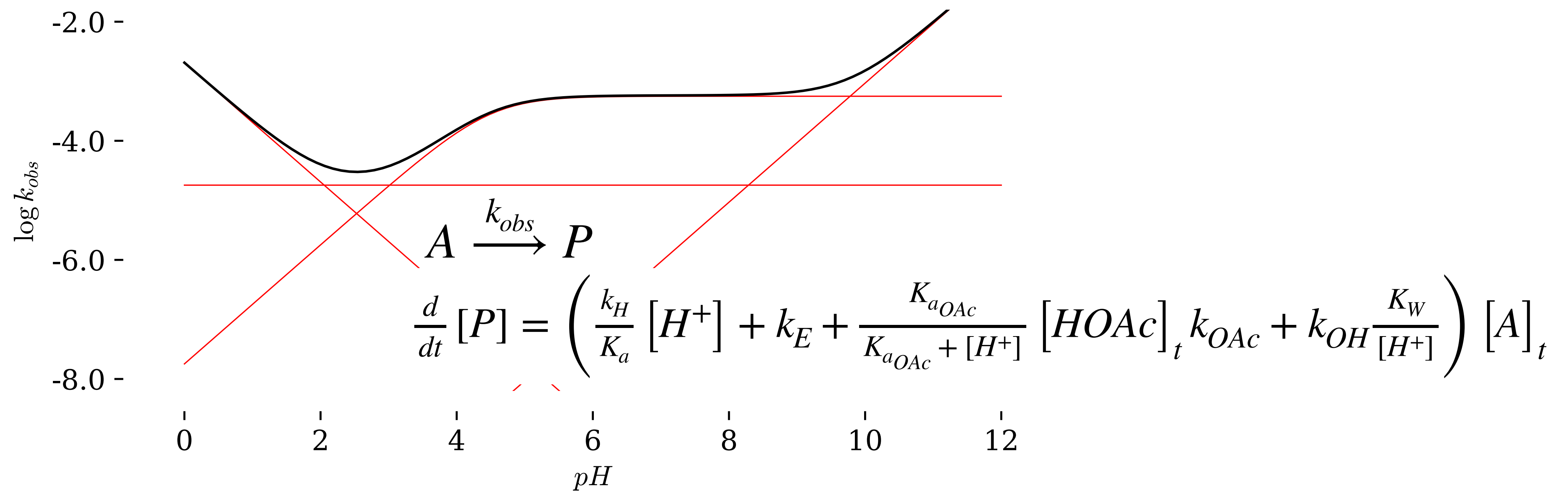
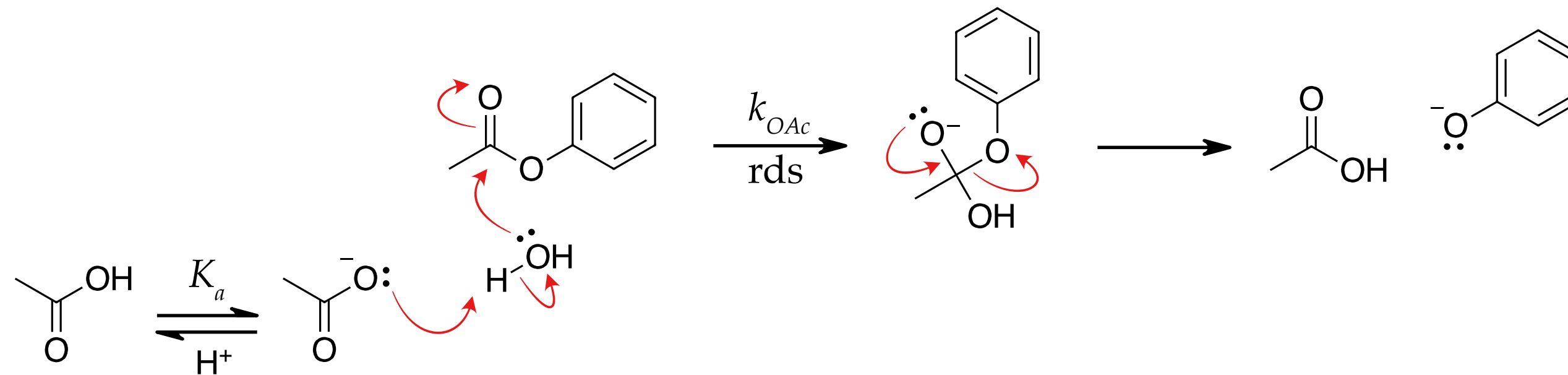


$$\frac{d}{dt} [P] = \left(\frac{k_H}{K_a} [H^+] + k_E + k_{OH} \frac{K_w}{[H^+]} \right) [A]_t$$

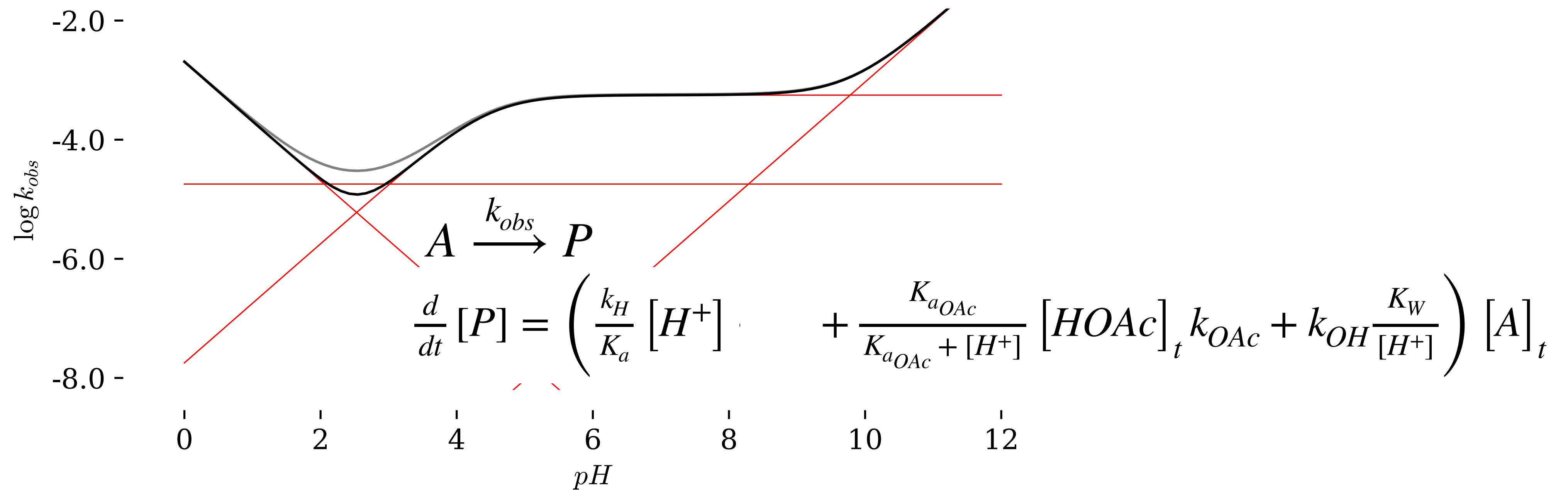
In Acetate Buffer



In Acetate Buffer

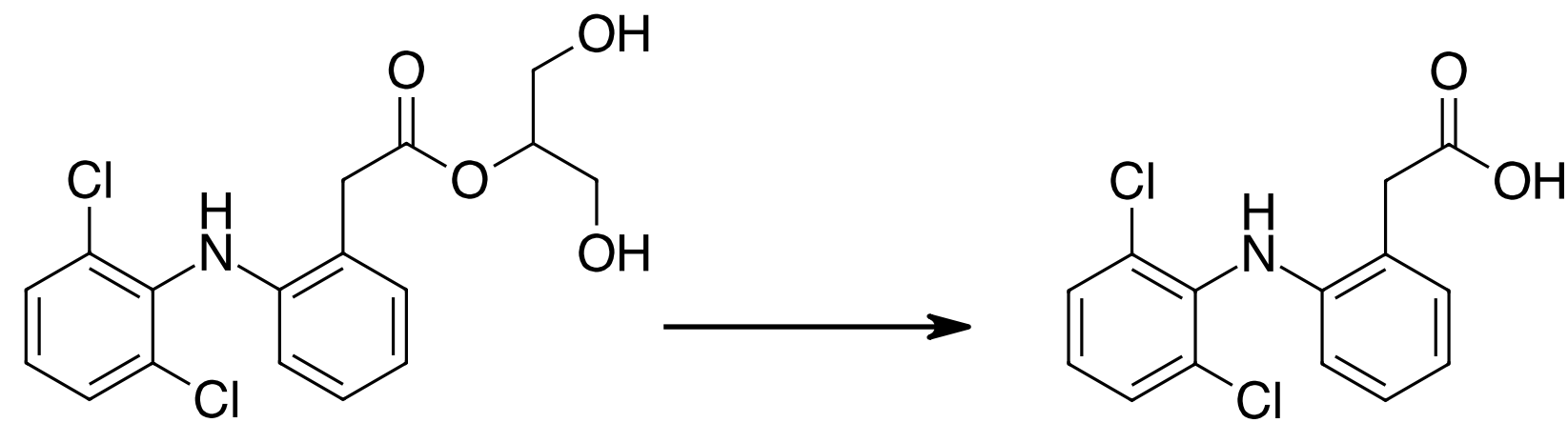


No Water Rate



Exploring an Experiment

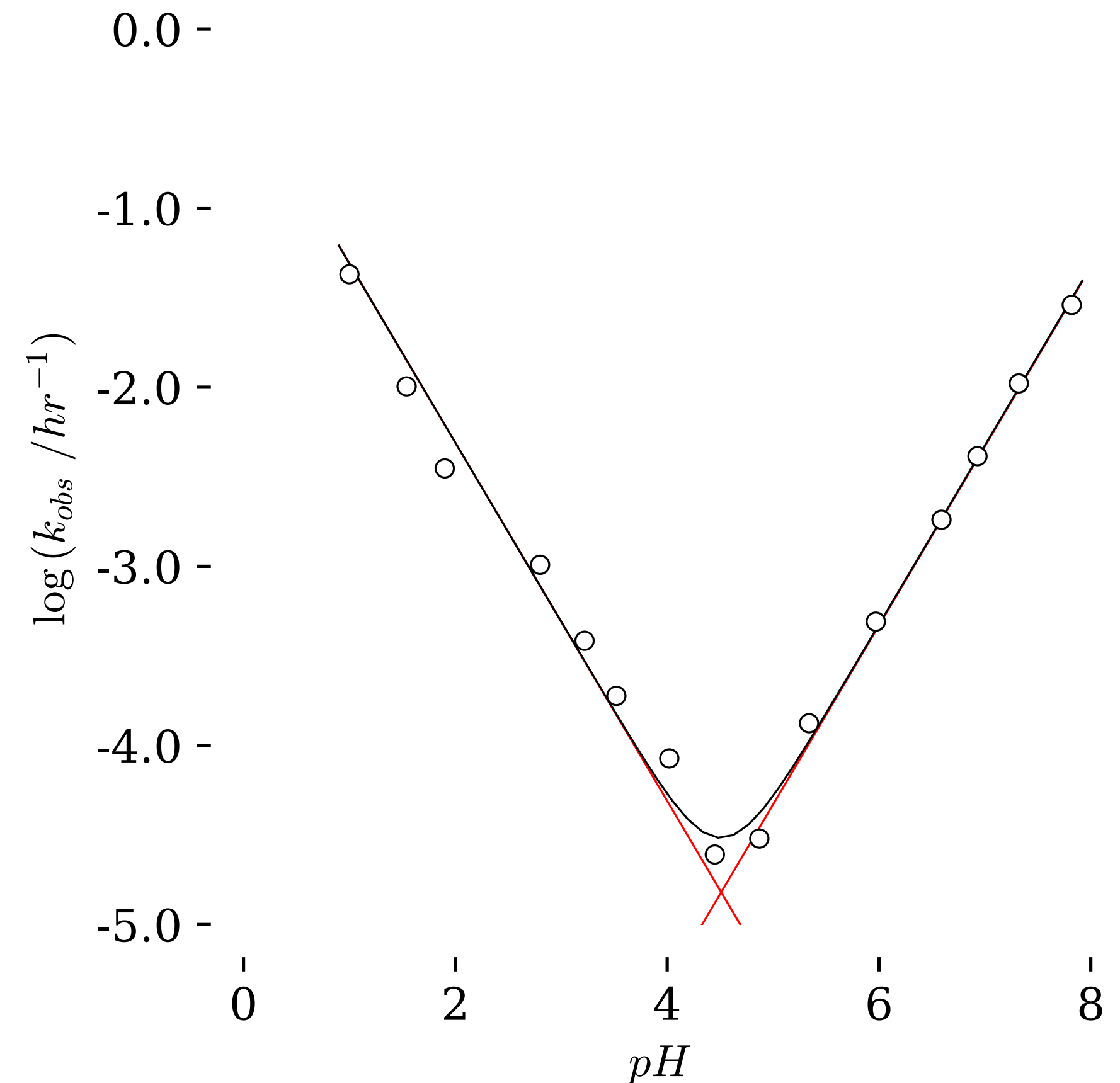
Wang, F.; Finnin, J; et al., "The Hydrolysis of Diclofenac Esters: Synthetic Prodrug Building Blocks for Biodegradable Drug-Polymer Conjugates."
J. Pharm. Sci., **2016**, 105, 773-785. <https://doi.org/10.1002/jps.24665>



$$\frac{d}{dt} [P] = \left(\frac{k_H}{K_a} [H^+] + k_{OH} \frac{K_W}{[H^+]} \right) [A]_t$$

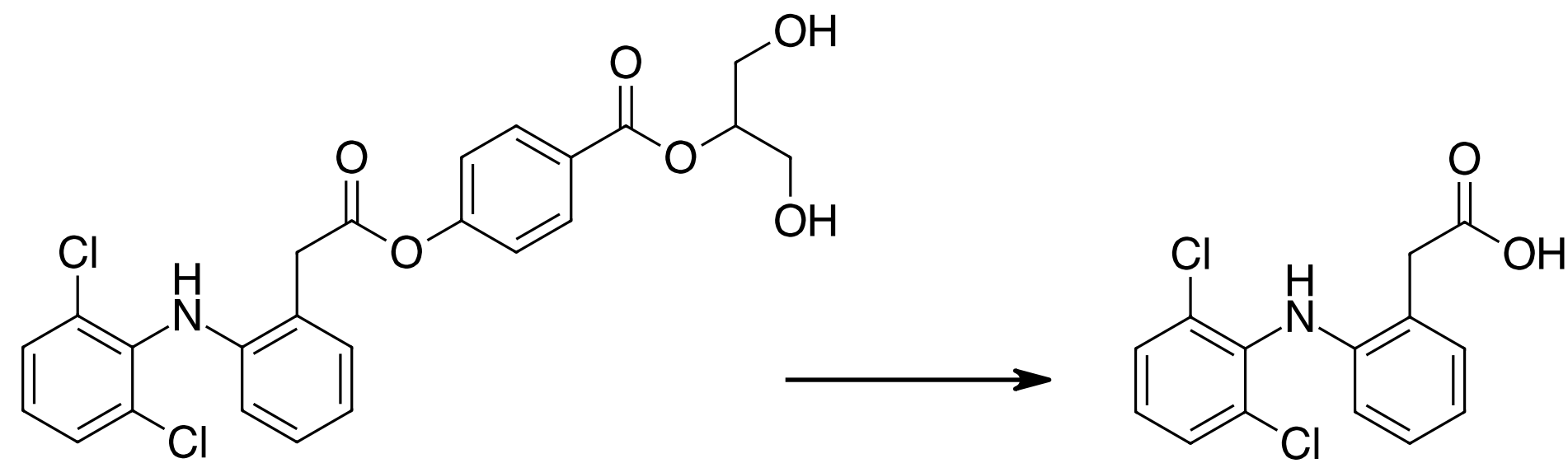
$$\frac{k_H}{K_a} = 0.49 \pm 0.05 \text{ M}^{-1} \text{ hr}^{-1}$$

$$k_{OH} = (1.97 \pm 0.22) \times 10^5 \text{ M}^{-1} \text{ hr}^{-1}$$



Exploring an Experiment

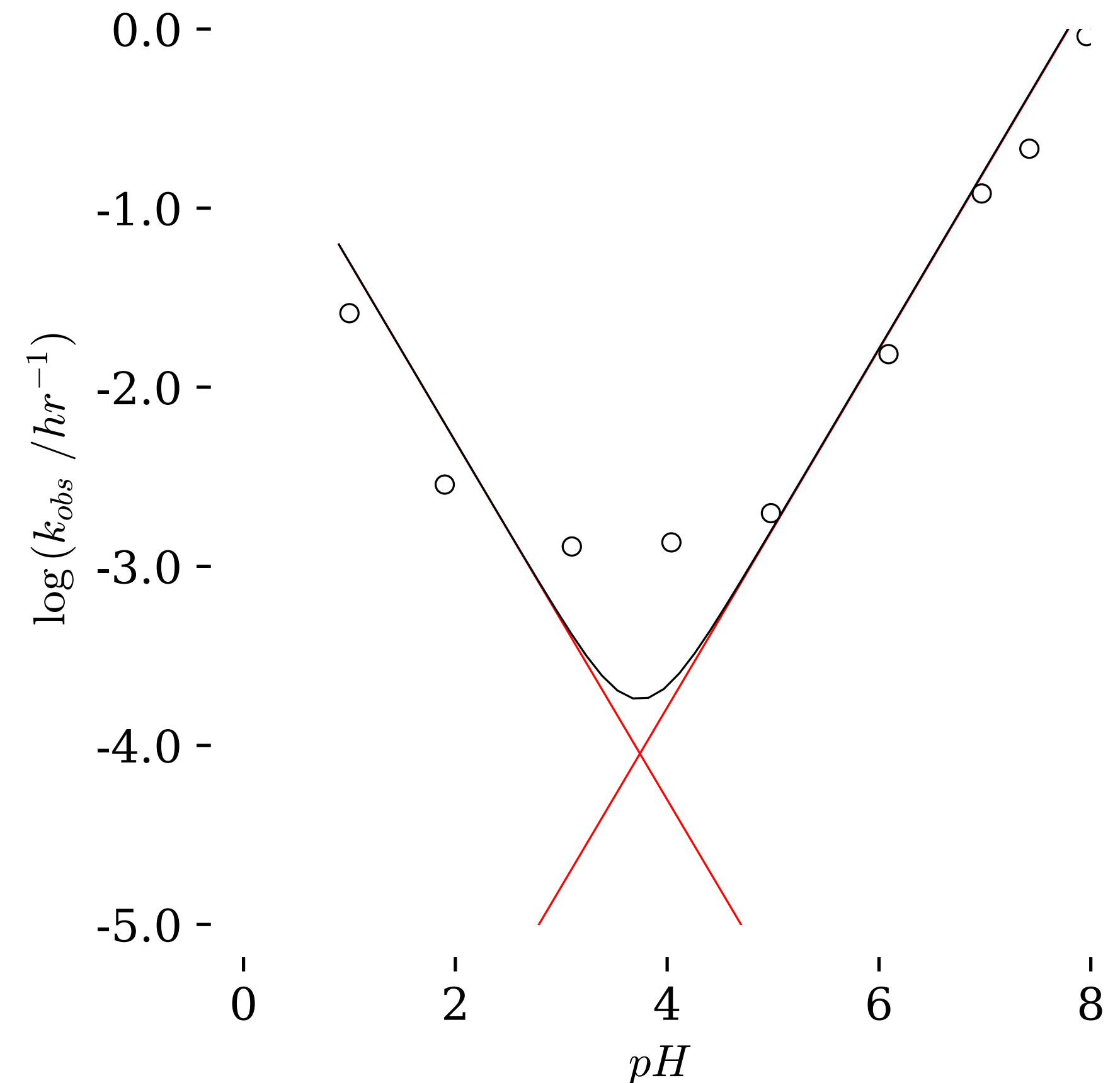
Wang, F.; Finnin, J; et al., "The Hydrolysis of Diclofenac Esters: Synthetic Prodrug Building Blocks for Biodegradable Drug-Polymer Conjugates."
J. Pharm. Sci., **2016**, 105, 773-785. <https://doi.org/10.1002/jps.24665>



$$\frac{d}{dt} [P] = \left(\frac{k_H}{K_a} [H^+] + k_{OH} \frac{K_W}{[H^+]} \right) [A]_t$$

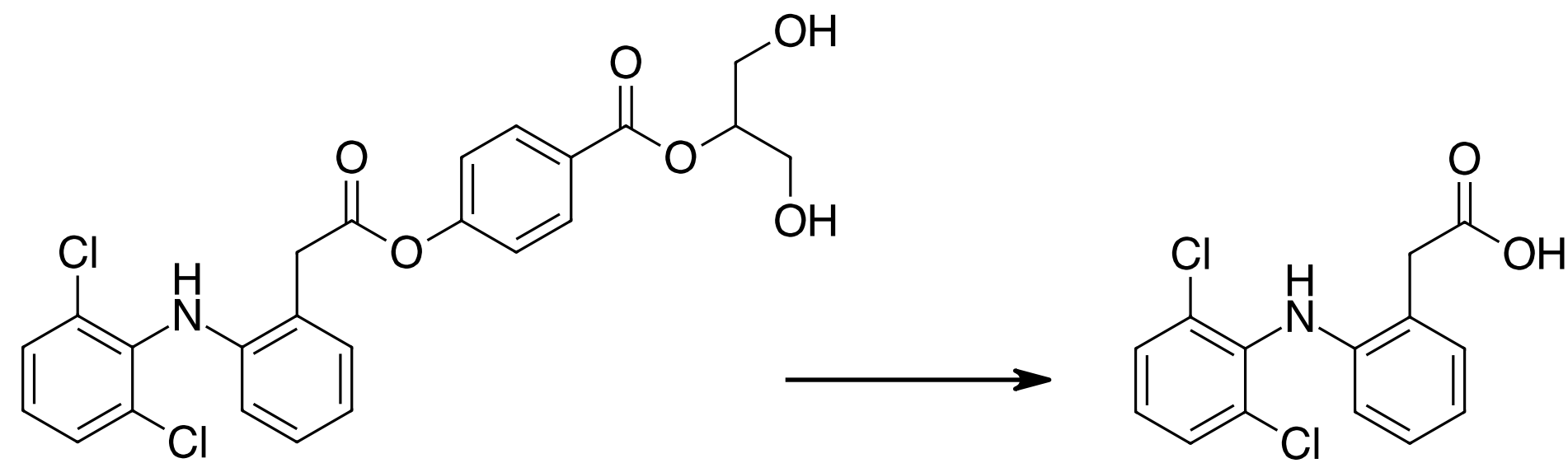
$$\frac{k_H}{K_a} = 0.49 \pm 0.28 \text{ M}^{-1} \text{ hr}^{-1}$$

$$k_{OH} = (1.62 \pm 0.66) \times 10^6 \text{ M}^{-1} \text{ hr}^{-1}$$



Exploring an Experiment

Wang, F.; Finnin, J; et al., "The Hydrolysis of Diclofenac Esters: Synthetic Prodrug Building Blocks for Biodegradable Drug-Polymer Conjugates."
J. Pharm. Sci., **2016**, 105, 773-785. <https://doi.org/10.1002/jps.24665>

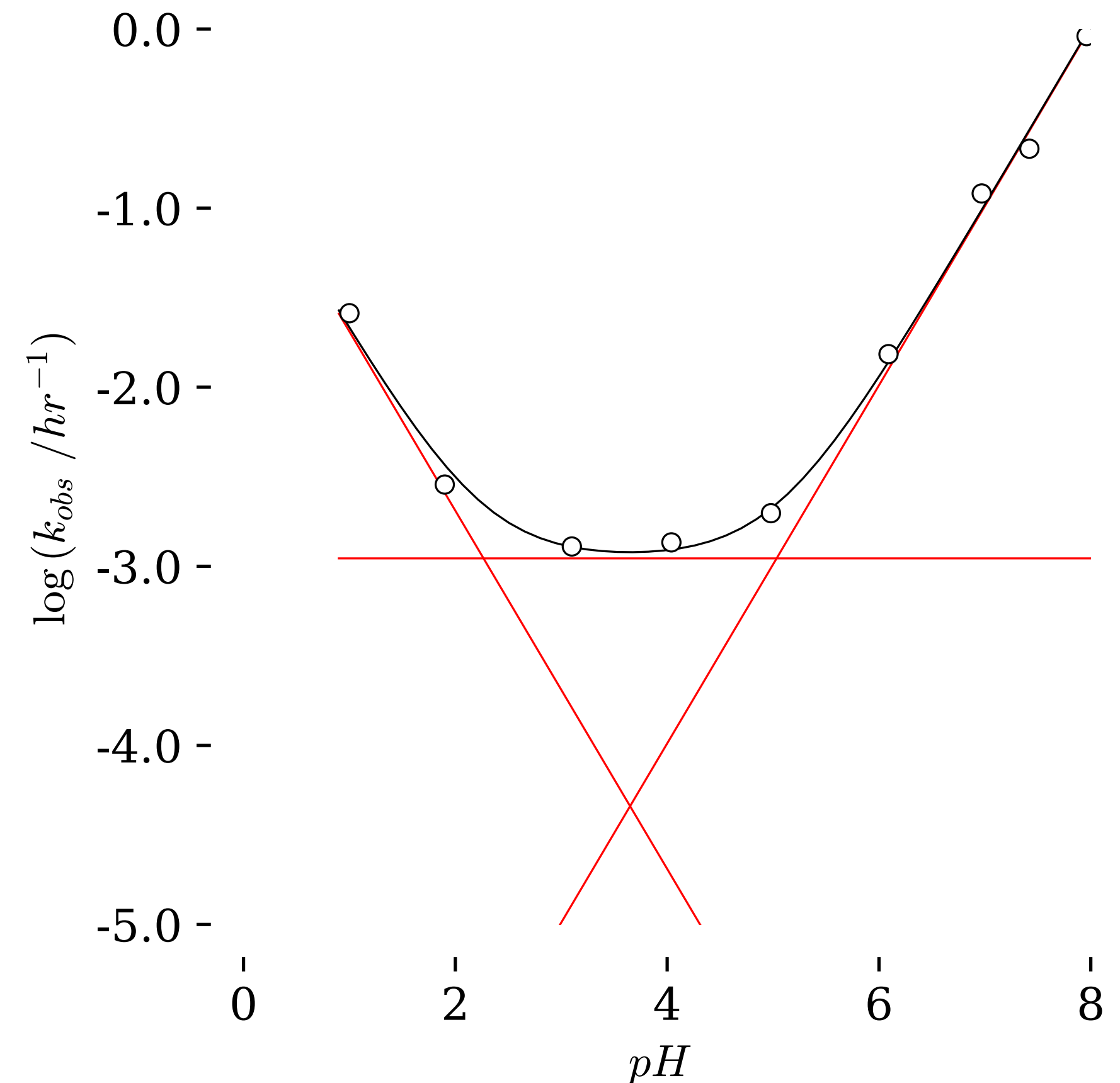


$$\frac{d}{dt} [P] = \left(\frac{k_H}{K_a} [H^+] + k_{H_2O} + k_{OH} \frac{K_w}{[H^+]} \right) [A]_t$$

$$\frac{k_H}{K_a} = 0.20 \pm 0.03 \text{ } M^{-1} hr^{-1}$$

$$k_{OH} = (1.02 \pm 0.10) \times 10^6 \text{ } M^{-1} hr^{-1}$$

$$k_{H_2O} = (1.11 \pm 0.06) \times 10^{-2} \text{ } hr^{-1}$$



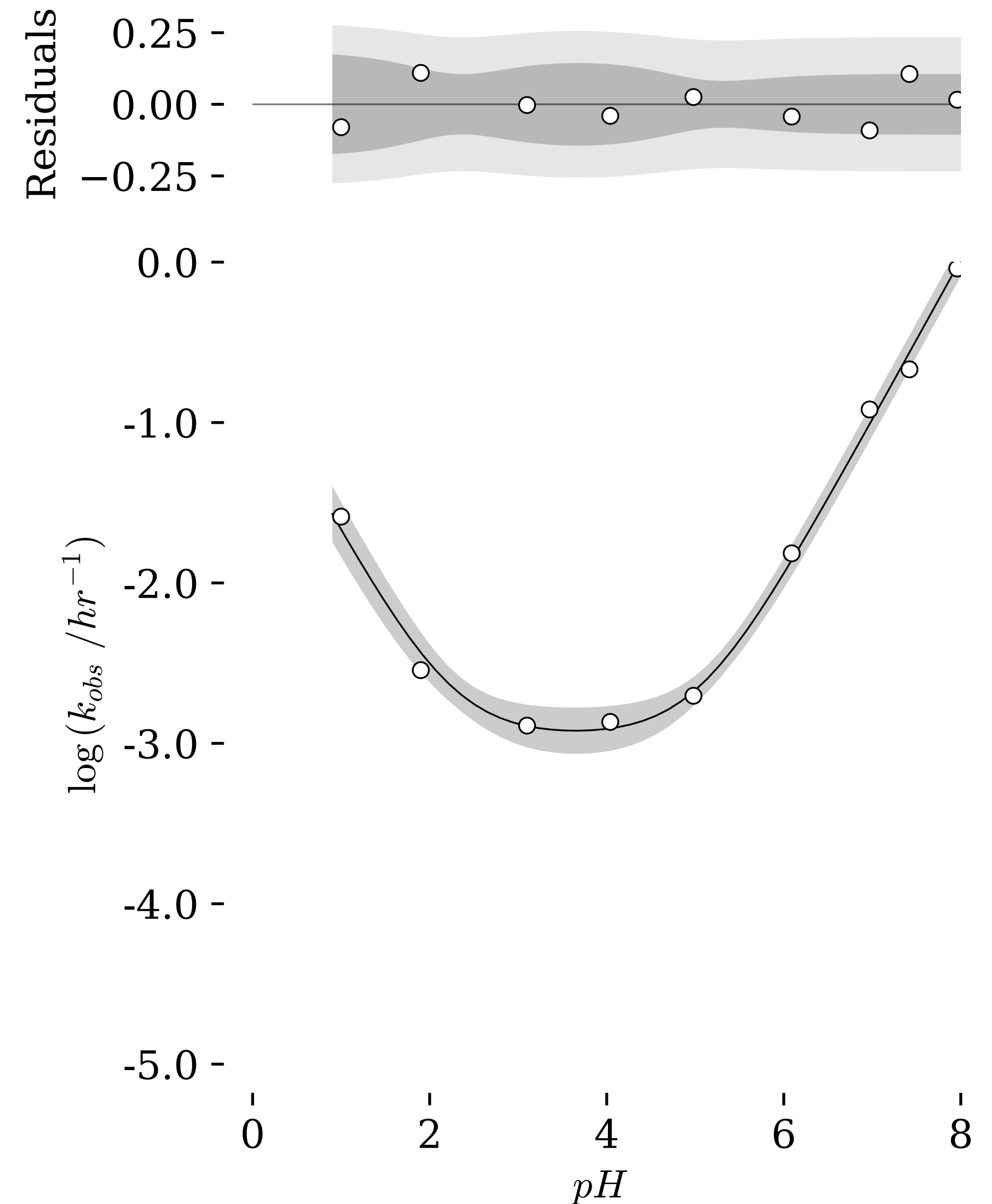
Exploring an Experiment

```
def profile3(pH, kHKa, kH2O, kOH):  
    H = 10**(-pH)  
    Kw = 10**(-13.995)  
    k1 = kHKa * H  
    k2 = kOH * Kw/H  
    kobs = k1 + k2 + kH2O  
    return np.log10(kobs)  
  
mod = lmfit.Model(profile3)  
  
pars = mod.make_params(kHKa = dict(value = 1, min=0),  
                       kH2O = dict(value = 1, min=0),  
                       kOH = dict(value = 1, min=0))  
  
x = df["pH"]  
y = np.log10(df["k1 (10-5 hr-1)"] * 1E-5)  
  
result = mod.fit(y, pars, pH=x)  
result.fit_report()
```

```
[[Model]]  
    Model(profile3)  
[[Fit Statistics]]  
    # fitting method      = leastsq  
    # function evals      = 38  
    # data points         = 9  
    # variables           = 3  
    chi-square            = 0.04226078  
    reduced chi-square    = 0.00704346  
    Akaike info crit      = -42.2500836  
    Bayesian info crit    = -41.6584099  
    R-squared             = 0.99524758  
[[Variables]]  
    kHKa:  0.20442919 +/- 0.03429308 (16.78%) (init = 1)  
    kH2O:  0.00110699 +/- 1.6074e-04 (14.52%) (init = 1)  
    kOH:   1024894.12 +/- 99733.5348 (9.73%) (init = 1)  
[[Correlations]] (unreported correlations are < 0.100)  
    C(kHKa, kH2O) = -0.2353  
    C(kH2O, kOH)  = -0.1581
```

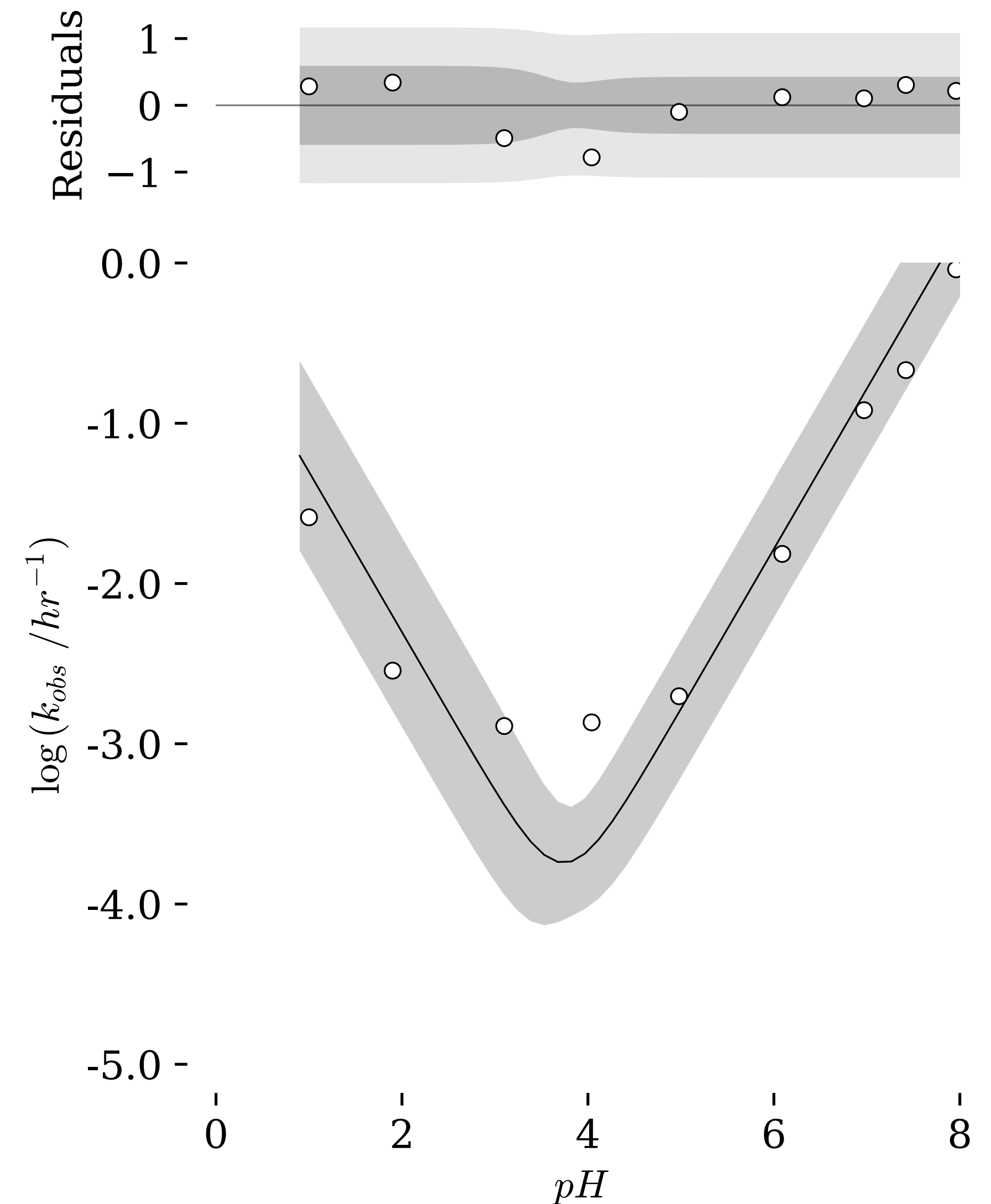
Exploring an Experiment

```
[[Model]]
  Model(profile3)
[[Fit Statistics]]
  # fitting method      = leastsq
  # function evals      = 38
  # data points         = 9
  # variables           = 3
  chi-square            = 0.04226078
  reduced chi-square    = 0.00704346
  Akaike info crit      = -42.2500836
  Bayesian info crit    = -41.6584099
  R-squared             = 0.99524758
[[Variables]]
  kHKa:  0.20442919 +/- 0.03429308 (16.78%) (init = 1)
  kH2O:  0.00110699 +/- 1.6074e-04 (14.52%) (init = 1)
  kOH:   1024894.12 +/- 99733.5348 (9.73%) (init = 1)
[[Correlations]] (unreported correlations are < 0.100)
  C(kHKa, kH2O) = -0.2353
  C(kH2O, kOH)  = -0.1581
```



Exploring an Experiment

```
[[Model]]
  Model(profile2)
[[Fit Statistics]]
  # fitting method      = leastsq
  # function evals      = 76
  # data points         = 9
  # variables           = 2
  chi-square            = 1.22109862
  reduced chi-square    = 0.17444266
  Akaike info crit      = -13.9772626
  Bayesian info crit    = -13.5828134
  R-squared             = 0.86268176
[[Variables]]
  kHka:  0.49762839 +/- 0.27935405 (56.14%) (init = 1)
  kOH:   1626118.32 +/- 659723.082 (40.57%) (init = 1)
```



Exploring an Experiment

```
[[Model]]
  Model(profile2)
[[Fit Statistics]]
  # fitting method      = leastsq
  # function evals      = 52
  # data points         = 15
  # variables           = 2
  chi-square            = 0.20984310
  reduced chi-square    = 0.01614178
  Akaike info crit     = -60.0416807
  Bayesian info crit   = -58.6255803
  R-squared             = 0.98618859
[[Variables]]
  kHka:  0.49008574 +/- 0.05380652 (10.98%) (init = 1)
  kOH:   46783.1142 +/- 5238.39283 (11.20%) (init = 1)
```

